NFH-Kuza-Google — Beckn Schema Definitions

Four generic extension schemas + `verticalAttributes` pattern, supporting any domain

BECKN DOMAINS	GENERIC SCHEMAS	SUB-SCHEMAS	VERTICALS DOCUMENTED
`local-services` · `health-services`	4	5	10 (extensible)

PROTOCOL	FORMAT
Beckn v2.0.0	OpenAPI 3.1

1. Overview

The Unified 4-Schema Model

This document defines **four generic, domain-agnostic extension schemas** that plug into the Beckn v2 protocol. Previously the protocol exposed 14 domain-specific schemas (7 for agri/rural, 7 for health); those have been collapsed into this single set of four:

SCHEMA	PREFIX	CONTAINER	PURPOSE
`ServiceItemAttributes`	`sia:`	`itemAttributes`	Describes the service / product / content / advisory on offer or being sought
`ServiceOfferAttributes`	`soa:`	`offerAttributes`	Pricing, availability, payment methods, cancellation policy
`ServiceFulfillmentAttributes`	`sfa:`	`fulfillmentAttributes`	Location, scheduling, delivery mode, lifecycle status
`ServiceOrderAttributes`	`sor:`	`orderAttributes`	Payment, feedback, outcome capture, consent & audit

The `verticalAttributes` Pattern

Every generic schema exposes an optional open-object field called **`verticalAttributes`**. This is a free-form key-value bag where a specific vertical (*poultry vaccination*, *TB screening*, *school fees*, etc.) can carry its domain-specific fields without requiring a schema change.

By convention, every `verticalAttributes` object MUST include:

- `vertical` REQUIRED — the vertical identifier, e.g. `"agri-livelihood-poultry"`, `"health-tb-screening"`, `"household-education"`.

The OpenAPI schema validates that `verticalAttributes` is an object. Field-level validation happens at the application layer, driven by the `vertical` value. The "Vertical Cookbook" (Section 6) documents the conventional shape for each known vertical.

> **Adding a new use case no longer requires a schema change.** To onboard a new vertical, simply document what `verticalAttributes` should contain. The four generic schemas stay stable; the Beckn protocol stays stable; only the application layer learns a new vertical-to-field mapping.

Currently Documented Verticals

`agri-livelihood-poultry`	`agri-livelihood-feed`	`household-education`
LOCAL-SERVICES	LOCAL-SERVICES	LOCAL-SERVICES

Vet services, vaccination, poultry health

Feed, inputs and consumables supply

School fees, admissions, tutoring

`household-health`
LOCAL-SERVICES
Household checkups, clinic visits

`market-livestock-products`
LOCAL-SERVICES
Egg / produce aggregation and sale

`advisory-agri`
LOCAL-SERVICES
Agri advisory content / training

`health-tb-screening`
HEALTH-SERVICES
TB symptom screening, GeneXpert, X-ray

`health-tb-treatment`
HEALTH-SERVICES
DSTB / DRTB / TPT regimens

`health-tb-supervision`
HEALTH-SERVICES
SCTLC / CTLC supervision visits

`advisory-health-tb`
HEALTH-SERVICES
TB awareness & adherence content

2. Beckn Domain Codes & Categories

The same four generic schemas are used for both Beckn domain codes. The `vertical` discriminator inside `verticalAttributes` tells the application layer which vertical-specific contract applies.

BECKN DOMAIN	SCOPE	VERTICALS
`local-services`	Rural livelihoods, household services, market aggregation, agri advisory	agri-livelihood-poultry, agri-livelihood-feed, household-education, household-health, market-livestock-products, advisory-agri
`health-services`	TB programme (TIBU integration): screening, treatment, supervision, post-TB care	health-tb-screening, health-tb-treatment, health-tb-supervision, advisory-health-tb

Category Codes — `local-services`

`Livelihood-Services`
LIVELIHOOD FLOW
Income / livelihood services and inputs (veterinary, crop, feed, trades)

`Household-Services`
HOUSEHOLD FLOW
Household needs (education, health, insurance, consumer goods)

`Market-Produce`
MARKET FLOW
Aggregation and sale of produce. Farmer is seller; provider is buyer/aggregator.

`Advisory-Content`
ADVISORY FLOW
Informational / training content. Free, discover-only — no order lifecycle.

Category Codes — `health-services`

`TB-Screening`
SCREENING FLOW
Symptom screening, sputum collection, X-ray, contact tracing (GeneXpert)

`TB-Treatment`
TREATMENT FLOW
DSTB / DRTB / TPT regimens with adherence tracking and ADR reporting

`TB-Supervision`
SUPERVISION FLOW
SCTLC / CTLC visits, data-quality audits, mortality audits, mentorship

Post-TB-Care

ADVISORY / FOLLOW-UP

TB awareness, adherence education, side-effects management content

3. Generic Schemas

Each of the four schemas below is domain-agnostic. Any vertical across any Beckn domain plugs into them; vertical-specific data lives inside the `verticalAttributes` bag.

3.1 ServiceItemAttributes

sia: **ServiceItemAttributes**

Container: **itemAttributes** Beckn entity: **Item / Resource** Applies to: **all verticals**

Describes the **service, product, content, or advisory** that is being offered or sought. Generic across every domain and vertical — from a vet vaccination visit, to a sputum collection, to an egg-aggregation offer, to an educational video.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object` {code, name}	REQUIRED	Domain grouping for the item	`{"code": "tb-screening", "name": "TB Screening"}`
`serviceName`	`string`	REQUIRED	Human-readable name	`"GeneXpert Sputum Test"`
`serviceDescription`	`string`	OPTIONAL	Free-text description	`"Collection point sputum testing, 48h turnaround"`
`serviceMode`	`enum`	REQUIRED	How the service is delivered. ON_SITE_VISIT FACILITY_VISIT PROVIDER_LOCATION COLLECTION_POINT DELIVERY DIGITAL REMOTE	`"FACILITY_VISIT"`
`quantity`	`QuantitySpec`	OPTIONAL	Quantity: birds, trays, doses, visits, sessions	`{"value": 50, "unit": "birds", "description": "Flock size"}`
`duration`	`string` (ISO 8601)	OPTIONAL	Estimated service duration	`"PT2H"`
`contentType`	`enum`	OPTIONAL	TEXT VIDEO PDF AUDIO IMAGE INFOGRAPHIC INTERACTIVE	`"VIDEO"`
`contentUrl`	`uri`	OPTIONAL	Resolvable URL for the content	`"https://cdn.example/tb-adherence.mp4"`

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`contentMimeType`	`string`	OPTIONAL	MIME type	`"video/mp4"`
`contentDuration`	`string` (ISO 8601)	OPTIONAL	Content play length	`"PT4M30S"`
`targetBeneficiary`	`string`	OPTIONAL	Who the service is for	`"Recently diagnosed TB patients"`
`documentsRequired`	`string[]`	OPTIONAL	Documents the seeker must bring	`["National ID", "Admission letter"]`
`frequency`	`enum`	OPTIONAL	ONE_TIME RECURRING SEASONAL	`"RECURRING"`
`requiredCapabilities`	`string[]`	OPTIONAL	Generic capability tags	`["GeneXpert", "Vet-certified"]`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1 language codes	`["en", "sw"]`
`tags`	`string[]`	OPTIONAL	Free-form tags	`["tb", "awareness"]`
`verticalAttributes`	`object`	OPTIONAL	Vertical-specific extension bag. See Section 6.	`{"vertical": "health-tb-screening", ...}`

OpenAPI Definition

```yaml
ServiceItemAttributes:
  type: object
  required: [subCategory, serviceName, serviceMode]
  properties:
    subCategory:
      type: object
      required: [code, name]
      properties:
        code: { type: string }
        name: { type: string }
    serviceName: { type: string }
    serviceDescription: { type: string }
    serviceMode:
      type: string
      enum: [ON_SITE_VISIT, FACILITY_VISIT, PROVIDER_LOCATION, COLLECTION_POINT, DELIVERY, DIGITAL, REMOTE]
    quantity: { $ref: "#/components/schemas/QuantitySpec" }
    duration: { type: string, format: duration }
    contentType:
      type: string
      enum: [TEXT, VIDEO, PDF, AUDIO, IMAGE, INFOGRAPHIC, INTERACTIVE]
    contentUrl: { type: string, format: uri }
    contentMimeType: { type: string }
    contentDuration: { type: string, format: duration }
    targetBeneficiary: { type: string }
    documentsRequired: { type: array, items: { type: string } }
    frequency:
      type: string
      enum: [ONE_TIME, RECURRING, SEASONAL]
    requiredCapabilities: { type: array, items: { type: string } }
    availableLanguages: { type: array, items: { type: string } }
    tags: { type: array, items: { type: string } }
    verticalAttributes:
      type: object
      additionalProperties: true
      required: [vertical]
```

```
      properties:
        vertical: { type: string }
```

Examples

Example 1 — Poultry vaccination (agri livelihood):

```
{
  "subCategory": { "code": "livestock-services", "name": "Livestock Services" },
  "serviceName": "Newcastle Vaccination",
  "serviceMode": "ON_SITE_VISIT",
  "quantity": { "value": 50, "unit": "birds", "description": "Flock size" },
  "duration": "PT1H",
  "requiredCapabilities": ["Vet-certified"],
  "verticalAttributes": {
    "vertical": "agri-livelihood-poultry",
    "livestockType": "Poultry",
    "breed": "Broiler",
    "vaccineType": "Newcastle",
    "ageGroup": "Adult"
  }
}
```

Example 2 — TB sputum screening (health):

```
{
  "subCategory": { "code": "tb-screening", "name": "TB Screening" },
  "serviceName": "GeneXpert Sputum Test",
  "serviceMode": "COLLECTION_POINT",
  "targetBeneficiary": "Adults with cough > 2 weeks",
  "documentsRequired": ["National ID"],
  "requiredCapabilities": ["GeneXpert"],
  "verticalAttributes": {
    "vertical": "health-tb-screening",
    "screeningType": "SPUTUM_COLLECTION",
    "diseaseTarget": "Tuberculosis",
    "diagnosticMethod": "GeneXpert",
    "expectedTurnaroundHours": 48,
    "isFreeAtPointOfCare": true
  }
}
```

Example 3 — Advisory video (agri content):

```
{
  "subCategory": { "code": "advisory-content", "name": "Advisory Content" },
  "serviceName": "Poultry Disease Prevention",
  "serviceMode": "DIGITAL",
  "contentType": "VIDEO",
  "contentUrl": "https://cdn.example/poultry-care.mp4",
  "contentMimeType": "video/mp4",
  "contentDuration": "PT6M",
  "availableLanguages": ["en", "sw"],
  "verticalAttributes": {
    "vertical": "advisory-agri",
    "topic": "Poultry Care",
    "subTopic": "Disease Prevention",
    "sourceProvider": "Google Content API"
  }
}
```

3.2 ServiceOfferAttributes

soa: **ServiceOfferAttributes**

Container: **offerAttributes** Beckn entity: **Offer**

Pricing, coverage, availability windows, accepted payment methods, capacity, and cancellation policy for an offered service.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`priceAmount`	`number`	OPTIONAL	Price. Use 0 for free services.	`300`
`currency`	`string` (ISO 4217)	OPTIONAL	Currency code	`"KES"`
`priceUnit`	`string`	OPTIONAL	What the price is per	`"per bird"`
`pricingModel`	`enum`	OPTIONAL	FREE FIXED PER_UNIT PER_VISIT NEGOTIABLE	`"PER_VISIT"`
`isFreeAtPointOfCare`	`boolean`	OPTIONAL	True for govt-funded / donor-funded services	`true`
`coveredBy`	`enum`	OPTIONAL	GOVERNMENT_PROGRAM INSURANCE OUT_OF_POCKET NGO_FUNDED GLOBAL_FUND NOT_APPLICABLE	`"GLOBAL_FUND"`
`coverageNote`	`string`	OPTIONAL	Free text about coverage	`"Free under NTLD"`
`acceptedPaymentMethods`	`enum[]`	OPTIONAL	CASH MPESA INSURANCE_CLAIM NONE	`["CASH", "MPESA"]`
`capacity`	`object`	OPTIONAL	`{value, unit, period}` — provider capacity	`{"value": 20, "unit": "visits", "period": "day"}`
`serviceAvailability`	`object`	OPTIONAL	`{availableDays, timeSlots}`	`{"availableDays": ["Mon","Wed"], "timeSlots": ["Morning"]}`
`cancellationPolicy`	`object`	OPTIONAL	`{isCancellable, windowHours, reasons}`	`{"isCancellable": true, "windowHours": 24}`
`validFrom`	`datetime`	OPTIONAL	Offer valid from	`"2026-04-01T00:00:00Z"`
`validTo`	`datetime`	OPTIONAL	Offer valid to	`"2026-12-31T23:59:59Z"`
`verticalAttributes`	`object`	OPTIONAL	Vertical-specific extension bag	`{"vertical": "market-livestock-products", ...}`

OpenAPI Definition

```yaml
ServiceOfferAttributes:
  type: object
```

```yaml
    properties:
      priceAmount: { type: number }
      currency: { type: string }
      priceUnit: { type: string }
      pricingModel:
        type: string
        enum: [FREE, FIXED, PER_UNIT, PER_VISIT, NEGOTIABLE]
      isFreeAtPointOfCare: { type: boolean }
      coveredBy:
        type: string
        enum: [GOVERNMENT_PROGRAM, INSURANCE, OUT_OF_POCKET, NGO_FUNDED, GLOBAL_FUND, NOT_APPLICABLE]
      coverageNote: { type: string }
      acceptedPaymentMethods:
        type: array
        items: { type: string, enum: [CASH, MPESA, INSURANCE_CLAIM, NONE] }
      capacity:
        type: object
        properties:
          value: { type: number }
          unit: { type: string }
          period: { type: string }
      serviceAvailability:
        type: object
        properties:
          availableDays: { type: array, items: { type: string } }
          timeSlots: { type: array, items: { type: string } }
      cancellationPolicy:
        type: object
        properties:
          isCancellable: { type: boolean }
          windowHours: { type: integer }
          reasons: { type: array, items: { type: string } }
      validFrom: { type: string, format: date-time }
      validTo: { type: string, format: date-time }
      verticalAttributes:
        type: object
        additionalProperties: true
        required: [vertical]
        properties:
          vertical: { type: string }
```

Examples

Example 1 — Paid vet visit:

```json
{
  "priceAmount": 500,
  "currency": "KES",
  "priceUnit": "per visit",
  "pricingModel": "PER_VISIT",
  "coveredBy": "OUT_OF_POCKET",
  "acceptedPaymentMethods": ["CASH", "MPESA"],
  "capacity": { "value": 8, "unit": "visits", "period": "day" },
  "serviceAvailability": { "availableDays": ["Mon", "Tue", "Wed"], "timeSlots": ["Morning"] },
  "cancellationPolicy": { "isCancellable": true, "windowHours": 12 }
}
```

Example 2 — Free TB screening under Global Fund:

```json
{
  "priceAmount": 0,
  "pricingModel": "FREE",
  "isFreeAtPointOfCare": true,
  "coveredBy": "GLOBAL_FUND",
```

```
  "coverageNote": "Free under NTLD programme",
  "acceptedPaymentMethods": ["NONE"],
  "capacity": { "value": 30, "unit": "tests", "period": "day" }
}
```

Example 3 — Egg aggregator offer:

```
{
  "priceAmount": 350,
  "currency": "KES",
  "priceUnit": "per tray",
  "pricingModel": "NEGOTIABLE",
  "acceptedPaymentMethods": ["MPESA"],
  "validFrom": "2026-04-01T00:00:00Z",
  "validTo": "2026-04-30T23:59:59Z",
  "verticalAttributes": {
    "vertical": "market-livestock-products",
    "produceCategory": "Eggs",
    "acceptedGrades": ["Grade A", "Grade B"],
    "minimumQuantity": 5
  }
}
```

3.3 ServiceFulfillmentAttributes

sfa: **ServiceFulfillmentAttributes**

Container: **fulfillmentAttributes** Beckn entity: **Performance / Fulfillment**

Covers **where** the service happens, **when** it is scheduled, **how** it is delivered, and **what lifecycle state** it is in.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION
`serviceArea`	`ServiceArea[]`	REQUIRED (PROVIDER)	Provider's registered coverage areas (counties / sub-counties)
`seekerLocation`	`SeekerLocation`	REQUIRED (SEEKER)	Farmer / patient / household location
`providerLocation`	`Location`	OPTIONAL	For facility-based services
`facilityType`	`string`	OPTIONAL	e.g. `"Vet Clinic"`, `"Health Centre"`, `"Collection Point"`
`requiredCapabilities`	`string[]`	OPTIONAL	Provider must have these to fulfil
`providerCredentialRequired`	`string`	OPTIONAL	e.g. `"Vet-certified"`, `"Clinical Officer"`, `"DRTB-trained"`
`scheduledDate`	`date`	OPTIONAL	Seeker's preferred date
`scheduledTimeSlot`	`string`	OPTIONAL	e.g. `"Morning"`, `"Afternoon"`
`confirmedVisitTime`	`datetime`	OPTIONAL	Provider's confirmed time

FIELD	TYPE	REQUIRED	DESCRIPTION
`actualServiceDate`	`date`	OPTIONAL	When the service was actually delivered
`deliveryMode`	`enum`	REQUIRED	ON_SITE FACILITY_VISIT PROVIDER_LOCATION COLLECTION_POINT HOME_VISIT TELE_HEALTH MOBILE DIGITAL
`serviceStatus`	`enum`	REQUIRED	NEED_REGISTERED SEARCHED SELECTED CONFIRMED DELIVERED RECORDED OUTCOME_RECORDED CANCELLED REJECTED LOST_TO_FOLLOWUP CLOSED
`providerNotes`	`string`	OPTIONAL	Free-text notes from provider
`followUpRequired`	`boolean`	OPTIONAL	Does this service need follow-up?
`referralRequired`	`boolean`	OPTIONAL	Needs referral to another provider
`referralTarget`	`object`	OPTIONAL	`{targetId, targetName, reason, urgency}`
`verticalAttributes`	`object`	OPTIONAL	Vertical-specific extension bag

OpenAPI Definition

```yaml
ServiceFulfillmentAttributes:
  type: object
  required: [deliveryMode, serviceStatus]
  properties:
    serviceArea:
      type: array
      items: { $ref: "#/components/schemas/ServiceArea" }
    seekerLocation: { $ref: "#/components/schemas/SeekerLocation" }
    providerLocation: { $ref: "#/components/schemas/Location" }
    facilityType: { type: string }
    requiredCapabilities: { type: array, items: { type: string } }
    providerCredentialRequired: { type: string }
    scheduledDate: { type: string, format: date }
    scheduledTimeSlot: { type: string }
    confirmedVisitTime: { type: string, format: date-time }
    actualServiceDate: { type: string, format: date }
    deliveryMode:
      type: string
      enum: [ON_SITE, FACILITY_VISIT, PROVIDER_LOCATION, COLLECTION_POINT, HOME_VISIT, TELE_HEALTH, MOBILE, DIGITAL]
    serviceStatus:
      type: string
      enum: [NEED_REGISTERED, SEARCHED, SELECTED, CONFIRMED, DELIVERED, RECORDED, OUTCOME_RECORDED, CANCELLED, REJECTED,
LOST_TO_FOLLOWUP, CLOSED]
    providerNotes: { type: string }
    followUpRequired: { type: boolean }
    referralRequired: { type: boolean }
    referralTarget:
      type: object
      properties:
        targetId: { type: string }
        targetName: { type: string }
        reason: { type: string }
        urgency: { type: string }
    verticalAttributes:
      type: object
      additionalProperties: true
      required: [vertical]
      properties:
        vertical: { type: string }
```

Examples

Example 1 — On-site vaccination visit:

```
{
  "seekerLocation": { "county": "Nakuru", "subCounty": "Naivasha", "ward": "Maai Mahiu" },
  "facilityType": "On-farm",
  "providerCredentialRequired": "Vet-certified",
  "scheduledDate": "2026-04-15",
  "scheduledTimeSlot": "Morning",
  "deliveryMode": "ON_SITE",
  "serviceStatus": "CONFIRMED"
}
```

Example 2 — TB screening at collection point:

```
{
  "seekerLocation": { "county": "Nairobi", "subCounty": "Kibra" },
  "providerLocation": { "name": "Kibra Health Centre", "gpsCoordinates": { "latitude": -1.312, "longitude": 36.783 } },
  "facilityType": "Health Centre",
  "providerCredentialRequired": "Clinical Officer",
  "deliveryMode": "COLLECTION_POINT",
  "serviceStatus": "DELIVERED",
  "actualServiceDate": "2026-04-12",
  "referralRequired": true,
  "referralTarget": { "targetName": "Mbagathi DRTB Centre", "reason": "Rifampicin resistance", "urgency": "HIGH" }
}
```

Example 3 — Supervision visit (provider side):

```
{
  "serviceArea": [{ "countyCode": "NBI", "countyName": "Nairobi", "subCounties": [{ "subCountyCode": "KBR",
"subCountyName": "Kibra" }] }],
  "facilityType": "Health Centre",
  "providerCredentialRequired": "SCTLC",
  "deliveryMode": "FACILITY_VISIT",
  "serviceStatus": "RECORDED",
  "verticalAttributes": { "vertical": "health-tb-supervision", "supervisionType": "SCTLC_VISIT" }
}
```

3.4 ServiceOrderAttributes

sor: ServiceOrderAttributes

Container: **orderAttributes** Beckn entity: **Contract / Order**

Captures payment settlement, bidirectional feedback, service outcome, consent, and adverse-event audit for the transaction.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION
`seekerUuid`	`string`	OPTIONAL	Generic unique ID — farmer ID / patient UUID / household ID

FIELD	TYPE	REQUIRED	DESCRIPTION
`providerCredentialId`	`string`	OPTIONAL	Provider's professional or business credential ID
`recordReference`	`string`	OPTIONAL	Reference to clinical / business record (e.g. FHIR resource URL)
`seekerPaymentMethod`	`enum`	OPTIONAL	CASH MPESA INSURANCE_CLAIM NONE
`seekerPaymentStatus`	`enum`	OPTIONAL	PAID NOT_PAID NOT_APPLICABLE
`providerPaymentReceived`	`boolean`	OPTIONAL	Provider confirms receipt
`providerAmountReceived`	`number`	OPTIONAL	Amount received
`paymentCurrency`	`string`	OPTIONAL	ISO 4217
`serviceConfirmedBySeeker`	`boolean`	OPTIONAL	Seeker confirms delivery
`providerFulfillmentStatus`	`enum`	OPTIONAL	COMPLETED PARTIAL NOT_COMPLETED
`outcomeRecorded`	`boolean`	OPTIONAL	Outcome has been captured
`outcomeType`	`string`	OPTIONAL	Generic — domain interprets (e.g. `"TB_POSITIVE"`, `"CURED"`, `"FOLLOW_UP_NEEDED"`)
`outcomeDate`	`date`	OPTIONAL	Date outcome was recorded
`outcomeNote`	`string`	OPTIONAL	Free text
`providerFeedback`	`Feedback`	OPTIONAL	Provider's feedback on seeker
`seekerFeedback`	`Feedback`	OPTIONAL	Seeker's feedback on provider
`followUpRequired`	`boolean`	OPTIONAL	Follow-up required
`followUpScheduledDate`	`date`	OPTIONAL	Scheduled follow-up date
`followUpNotes`	`string`	OPTIONAL	Follow-up plan notes
`consentGiven`	`boolean`	OPTIONAL	Explicit consent for data sharing
`consentTimestamp`	`datetime`	OPTIONAL	When consent was captured
`adverseEventReported`	`boolean`	OPTIONAL	Any adverse event reported
`adverseEventDetails`	`object`	OPTIONAL	`{eventType, severity, onsetDate, action, outcome}`
`verticalAttributes`	`object`	OPTIONAL	Vertical-specific extension bag

OpenAPI Definition

```yaml
ServiceOrderAttributes:
  type: object
  properties:
    seekerUuid: { type: string }
    providerCredentialId: { type: string }
    recordReference: { type: string }
    seekerPaymentMethod:
      type: string
      enum: [CASH, MPESA, INSURANCE_CLAIM, NONE]
    seekerPaymentStatus:
      type: string
```

```yaml
        enum: [PAID, NOT_PAID, NOT_APPLICABLE]
      providerPaymentReceived: { type: boolean }
      providerAmountReceived: { type: number }
      paymentCurrency: { type: string }
      serviceConfirmedBySeeker: { type: boolean }
      providerFulfillmentStatus:
        type: string
        enum: [COMPLETED, PARTIAL, NOT_COMPLETED]
      outcomeRecorded: { type: boolean }
      outcomeType: { type: string }
      outcomeDate: { type: string, format: date }
      outcomeNote: { type: string }
      providerFeedback: { $ref: "#/components/schemas/Feedback" }
      seekerFeedback: { $ref: "#/components/schemas/Feedback" }
      followUpRequired: { type: boolean }
      followUpScheduledDate: { type: string, format: date }
      followUpNotes: { type: string }
      consentGiven: { type: boolean }
      consentTimestamp: { type: string, format: date-time }
      adverseEventReported: { type: boolean }
      adverseEventDetails:
        type: object
        properties:
          eventType: { type: string }
          severity: { type: string }
          onsetDate: { type: string, format: date }
          action: { type: string }
          outcome: { type: string }
      verticalAttributes:
        type: object
        additionalProperties: true
        required: [vertical]
        properties:
          vertical: { type: string }
```

Examples

Example 1 — Paid vet visit closed out:

```json
{
  "seekerUuid": "farmer-8812",
  "providerCredentialId": "KVB-2019-0042",
  "seekerPaymentMethod": "MPESA",
  "seekerPaymentStatus": "PAID",
  "providerPaymentReceived": true,
  "providerAmountReceived": 500,
  "paymentCurrency": "KES",
  "serviceConfirmedBySeeker": true,
  "providerFulfillmentStatus": "COMPLETED",
  "outcomeRecorded": true,
  "outcomeType": "VACCINATION_COMPLETED",
  "seekerFeedback": { "rating": 5, "additionalText": "Very punctual" }
}
```

Example 2 — TB screening outcome:

```json
{
  "seekerUuid": "TIBU-PAT-554321",
  "providerCredentialId": "KMPDC-CO-4412",
  "recordReference": "fhir:DiagnosticReport/tb-554321-001",
  "seekerPaymentMethod": "NONE",
  "providerFulfillmentStatus": "COMPLETED",
  "outcomeRecorded": true,
  "outcomeType": "TB_POSITIVE",
```

```
  "outcomeDate": "2026-04-14",
  "outcomeNote": "Rifampicin resistance detected; refer DRTB",
  "consentGiven": true,
  "consentTimestamp": "2026-04-12T09:15:00+03:00",
  "followUpRequired": true,
  "followUpScheduledDate": "2026-04-16"
}
```

Example 3 — TB treatment with adverse event:

```
{
  "seekerUuid": "TIBU-PAT-554321",
  "outcomeRecorded": true,
  "outcomeType": "TREATMENT_COMPLETED",
  "outcomeDate": "2026-10-12",
  "adverseEventReported": true,
  "adverseEventDetails": {
    "eventType": "Hepatotoxicity",
    "severity": "MODERATE",
    "onsetDate": "2026-06-22",
    "action": "Regimen modified; LFT monitored",
    "outcome": "Resolved"
  }
}
```

4. Sub-Schemas

These shared sub-types are referenced by the four generic schemas above.

ServiceArea

FIELD	TYPE	REQUIRED	DESCRIPTION
`countyCode`	`string`	REQUIRED	County code
`countyName`	`string`	REQUIRED	County name
`subCounties`	`array`	OPTIONAL	List of `{subCountyCode, subCountyName}`

SeekerLocation

FIELD	TYPE	REQUIRED	DESCRIPTION
`county`	`string`	REQUIRED	County name
`subCounty`	`string`	REQUIRED	Sub-county name
`ward`	`string`	OPTIONAL	Ward
`village`	`string`	OPTIONAL	Village
`gpsCoordinates`	`object`	OPTIONAL	`{latitude, longitude}`

Location

Generic location object used for provider / facility location. Fields: `name`, `address`, `county`, `subCounty`, `gpsCoordinates: {latitude, longitude}`. All fields optional; at least one of `address` or `gpsCoordinates` should be present.

Feedback

FIELD	TYPE	REQUIRED	DESCRIPTION
`rating`	`integer`	OPTIONAL	Rating 1-5
`selectedOption`	`string`	OPTIONAL	Selected option (e.g. *Very satisfied*)
`additionalText`	`string`	OPTIONAL	Free text

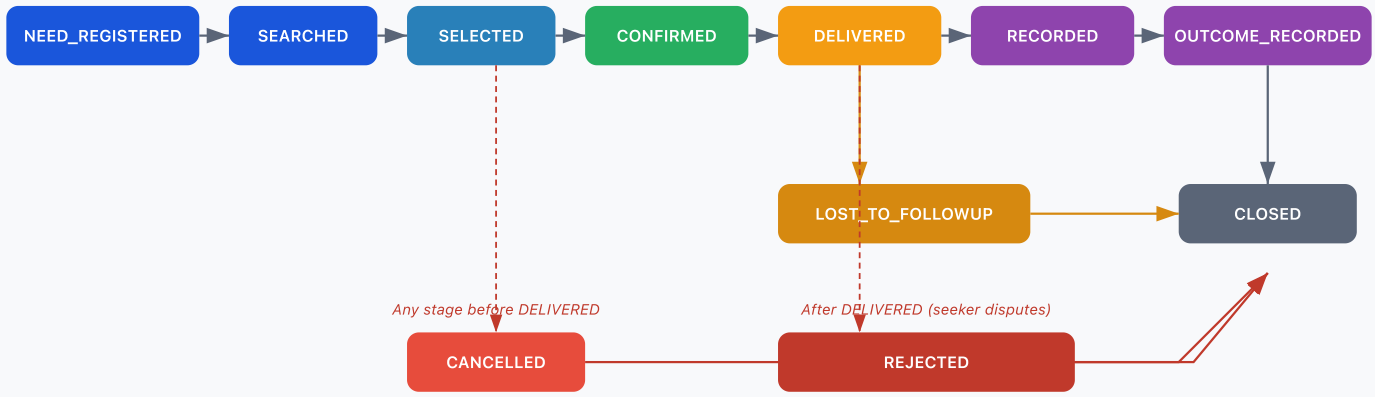
QuantitySpec

FIELD	TYPE	REQUIRED	DESCRIPTION
`value`	`number`	REQUIRED	Numeric value
`unit`	`string`	REQUIRED	Unit (birds, trays, doses, kg, visits, ...)
`description`	`string`	OPTIONAL	Free-text description

5. Service Status Lifecycle

The `serviceStatus` field in `ServiceFulfillmentAttributes` follows a **unified 8-stage lifecycle** that applies universally, regardless of Beckn domain or vertical. Advisory content is the only exception — it has no lifecycle and is delivered directly in `on_discover`.

Unified Service Status Lifecycle — applies to every domain & vertical



Lifecycle notes

The happy path has 8 stages: NEED_REGISTERED → SEARCHED → SELECTED → CONFIRMED → DELIVERED → RECORDED → OUTCOME_RECORDED → CLOSED.

Exception branches: CANCELLED (before DELIVERED), REJECTED (after DELIVERED), LOST_TO_FOLLOWUP (when seeker cannot be contacted).

All three exception states terminate in CLOSED.

Applies to every vertical. Advisory content bypasses the lifecycle entirely.

State transition rules:

FROM	TO	TRIGGERED BY
`NEED_REGISTERED`	`SEARCHED`	BAP issues discover; catalog returned
`SEARCHED`	`SELECTED`	Seeker selects a provider/item
`SELECTED`	`CONFIRMED`	Provider accepts the order
`CONFIRMED`	`DELIVERED`	Provider delivers the service
`DELIVERED`	`RECORDED`	Provider records completion (notes, data)
`RECORDED`	`OUTCOME_RECORDED`	Outcome captured (seeker confirmation OR clinical result OR sale completion)
`DELIVERED` / `RECORDED`	`LOST_TO_FOLLOWUP`	Seeker cannot be contacted
`OUTCOME_RECORDED` / `LOST_TO_FOLLOWUP`	`CLOSED`	Transaction finalized
Any stage before `DELIVERED`	`CANCELLED`	Either party cancels
After `DELIVERED`	`REJECTED`	Seeker disputes delivered service
`CANCELLED` / `REJECTED`	`CLOSED`	Automatic terminal closure

> **Advisory content has no lifecycle.** Any vertical whose `subCategory.code` is `advisory-content` (or equivalent) is delivered immediately via `contentUrl` in the `on_discover` response.

6. Vertical Cookbook

For each documented vertical, the cookbook specifies: (a) the Beckn domain, (b) the conventional `verticalAttributes` shape, and (c) a worked example showing the four generic schemas plugged together in a single Beckn message.

LOCAL-SERVICES **`agri-livelihood-poultry`**

Veterinary services, vaccinations and general poultry health for smallholder flocks. Mapped to category `Livelihood-Services`, sub-category `livestock-services`.

verticalAttributes shape

```
{
  "vertical": "agri-livelihood-poultry",
  "livestockType": "Poultry",
  "breed": "Broiler",
  "vaccineType": "Newcastle",
  "ageGroup": "Adult"
}
```

Full Beckn message (all 4 schemas)

```
{
  "context": { "domain": "local-services", "action": "on_confirm", "version": "2.0.0" },
  "itemAttributes": {
```

```
    "subCategory": { "code": "livestock-services", "name": "Livestock Services" },
    "serviceName": "Newcastle Vaccination", "serviceMode": "ON_SITE_VISIT",
    "quantity": { "value": 50, "unit": "birds" },
    "verticalAttributes": { "vertical": "agri-livelihood-poultry", "livestockType": "Poultry", "breed": "Broiler",
"vaccineType": "Newcastle" }
  },
  "offerAttributes": {
    "priceAmount": 500, "currency": "KES", "pricingModel": "PER_VISIT",
    "acceptedPaymentMethods": ["CASH", "MPESA"]
  },
  "fulfillmentAttributes": {
    "seekerLocation": { "county": "Nakuru", "subCounty": "Naivasha" },
    "providerCredentialRequired": "Vet-certified",
    "scheduledDate": "2026-04-15", "deliveryMode": "ON_SITE", "serviceStatus": "CONFIRMED"
  },
  "orderAttributes": {
    "seekerUuid": "farmer-8812", "providerCredentialId": "KVB-2019-0042",
    "seekerPaymentMethod": "MPESA", "seekerPaymentStatus": "PAID"
  }
}
```

LOCAL-SERVICES **agri-livelihood-feed**

Feed, inputs and consumables supply to smallholder farmers.

verticalAttributes shape

```
{
  "vertical": "agri-livelihood-feed",
  "inputName": "Broiler Feed",
  "brand": "Unga",
  "bagSize": "50kg"
}
```

Full Beckn message

```
{
  "context": { "domain": "local-services", "action": "on_confirm" },
  "itemAttributes": {
    "subCategory": { "code": "input-supply", "name": "Input Supply" },
    "serviceName": "Broiler Feed 50kg", "serviceMode": "DELIVERY",
    "quantity": { "value": 4, "unit": "bags" },
    "verticalAttributes": { "vertical": "agri-livelihood-feed", "inputName": "Broiler Feed", "brand": "Unga",
"bagSize": "50kg" }
  },
  "offerAttributes": { "priceAmount": 3200, "currency": "KES", "priceUnit": "per bag", "pricingModel": "PER_UNIT"
},
  "fulfillmentAttributes": { "seekerLocation": { "county": "Nakuru", "subCounty": "Naivasha" }, "deliveryMode":
"HOME_VISIT", "serviceStatus": "CONFIRMED" },
  "orderAttributes": { "seekerUuid": "farmer-8812", "seekerPaymentMethod": "MPESA", "seekerPaymentStatus": "PAID"
}
}
```

LOCAL-SERVICES **household-education**

School fee payment, admissions support, tutoring.

verticalAttributes shape

```
{
  "vertical": "household-education",
  "schoolName": "St. Mary Primary",
  "term": "Term 2 2026",
  "classLevel": "Grade 5"
}
```

Full Beckn message

```
{
  "context": { "domain": "local-services", "action": "on_confirm" },
  "itemAttributes": {
    "subCategory": { "code": "education", "name": "Education" },
    "serviceName": "Term Fee Payment", "serviceMode": "DIGITAL",
    "documentsRequired": ["School admission letter"],
    "verticalAttributes": { "vertical": "household-education", "schoolName": "St. Mary Primary", "term": "Term 2
2026", "classLevel": "Grade 5" }
  },
  "offerAttributes": { "priceAmount": 12000, "currency": "KES", "pricingModel": "FIXED", "acceptedPaymentMethods":
["MPESA"] },
  "fulfillmentAttributes": { "seekerLocation": { "county": "Nakuru", "subCounty": "Naivasha" }, "deliveryMode":
"DIGITAL", "serviceStatus": "DELIVERED" },
  "orderAttributes": { "seekerUuid": "household-0054", "seekerPaymentMethod": "MPESA", "seekerPaymentStatus":
"PAID", "providerAmountReceived": 12000 }
}
```

LOCAL-SERVICES **`household-health`**

Generic household health services: general checkups, clinic visits not scoped to a disease programme.

verticalAttributes shape

```
{
  "vertical": "household-health",
  "checkupType": "General",
  "facilityName": "Naivasha District Hospital"
}
```

Full Beckn message

```
{
  "context": { "domain": "local-services", "action": "on_confirm" },
  "itemAttributes": {
    "subCategory": { "code": "health", "name": "Health" },
    "serviceName": "General Checkup", "serviceMode": "FACILITY_VISIT",
    "verticalAttributes": { "vertical": "household-health", "checkupType": "General", "facilityName": "Naivasha
District Hospital" }
  },
  "offerAttributes": { "priceAmount": 800, "currency": "KES", "pricingModel": "FIXED" },
  "fulfillmentAttributes": { "seekerLocation": { "county": "Nakuru", "subCounty": "Naivasha" }, "facilityType":
"District Hospital", "deliveryMode": "FACILITY_VISIT", "serviceStatus": "CONFIRMED" },
```

```
  "orderAttributes": { "seekerUuid": "household-0054", "seekerPaymentMethod": "CASH" }
}
```

LOCAL-SERVICES **`market-livestock-products`**

Farmer-to-aggregator sale of eggs, poultry or other livestock products. The *seeker* is the farmer and the *provider* is the buying aggregator.

verticalAttributes shape

```
{
  "vertical": "market-livestock-products",
  "produceCategory": "Eggs",
  "acceptedGrades": ["Grade A", "Grade B"],
  "unitOfMeasure": "Tray (30 eggs)",
  "minimumQuantity": 5,
  "collectionMode": "FARM_GATE_PICKUP",
  "paymentTimeline": "UPON_COLLECTION"
}
```

Full Beckn message

```
{
  "context": { "domain": "local-services", "action": "on_confirm" },
  "itemAttributes": {
    "subCategory": { "code": "market-produce", "name": "Market Produce" },
    "serviceName": "Egg Aggregation", "serviceMode": "COLLECTION_POINT",
    "quantity": { "value": 10, "unit": "trays" },
    "verticalAttributes": { "vertical": "market-livestock-products", "produceCategory": "Eggs", "acceptedGrades":
["Grade A"] }
  },
  "offerAttributes": {
    "priceAmount": 350, "currency": "KES", "priceUnit": "per tray", "pricingModel": "NEGOTIABLE",
    "verticalAttributes": { "vertical": "market-livestock-products", "collectionMode": "FARM_GATE_PICKUP",
"paymentTimeline": "UPON_COLLECTION" }
  },
  "fulfillmentAttributes": { "seekerLocation": { "county": "Nakuru", "subCounty": "Naivasha" }, "deliveryMode":
"COLLECTION_POINT", "serviceStatus": "DELIVERED" },
  "orderAttributes": { "seekerUuid": "farmer-8812", "providerPaymentReceived": true, "providerAmountReceived":
3500, "paymentCurrency": "KES" }
}
```

LOCAL-SERVICES **`advisory-agri`**

Free agricultural advisory content (video, audio, text). Discover-only; no order lifecycle.

verticalAttributes shape

```
{
  "vertical": "advisory-agri",
  "topic": "Poultry Care",
  "subTopic": "Disease Prevention",
```

```
  "sourceProvider": "Google Content API"
}
```

Full Beckn message (on_discover)

```
{
  "context": { "domain": "local-services", "action": "on_discover" },
  "itemAttributes": {
    "subCategory": { "code": "advisory-content", "name": "Advisory Content" },
    "serviceName": "Poultry Disease Prevention", "serviceMode": "DIGITAL",
    "contentType": "VIDEO", "contentUrl": "https://cdn.example/poultry-care.mp4", "contentDuration": "PT6M",
    "availableLanguages": ["en", "sw"],
    "verticalAttributes": { "vertical": "advisory-agri", "topic": "Poultry Care", "subTopic": "Disease
Prevention", "sourceProvider": "Google Content API" }
  },
  "offerAttributes": { "priceAmount": 0, "pricingModel": "FREE" },
  "fulfillmentAttributes": { "deliveryMode": "DIGITAL", "serviceStatus": "DELIVERED" }
}
```

HEALTH-SERVICES **`health-tb-screening`**

TB symptom screening, sputum collection, chest X-ray, contact tracing. GeneXpert is the dominant diagnostic method.

verticalAttributes shape

```
{
  "vertical": "health-tb-screening",
  "screeningType": "SPUTUM_COLLECTION",
  "diseaseTarget": "Tuberculosis",
  "diagnosticMethod": "GeneXpert",
  "expectedTurnaroundHours": 48,
  "isFreeAtPointOfCare": true,
  "patientEligibility": { "minAge": 5, "riskFactors": ["HIV+", "Cough > 2 weeks"] }
}
```

Full Beckn message

```
{
  "context": { "domain": "health-services", "action": "on_confirm" },
  "itemAttributes": {
    "subCategory": { "code": "tb-screening", "name": "TB Screening" },
    "serviceName": "GeneXpert Sputum Test", "serviceMode": "COLLECTION_POINT",
    "documentsRequired": ["National ID"], "requiredCapabilities": ["GeneXpert"],
    "verticalAttributes": { "vertical": "health-tb-screening", "screeningType": "SPUTUM_COLLECTION",
"diseaseTarget": "Tuberculosis", "diagnosticMethod": "GeneXpert", "expectedTurnaroundHours": 48 }
  },
  "offerAttributes": { "priceAmount": 0, "pricingModel": "FREE", "isFreeAtPointOfCare": true, "coveredBy":
"GLOBAL_FUND" },
  "fulfillmentAttributes": { "seekerLocation": { "county": "Nairobi", "subCounty": "Kibra" }, "facilityType":
"Health Centre", "providerCredentialRequired": "Clinical Officer", "deliveryMode": "COLLECTION_POINT",
"serviceStatus": "DELIVERED" },
  "orderAttributes": { "seekerUuid": "TIBU-PAT-554321", "recordReference": "fhir:DiagnosticReport/tb-554321-001",
"outcomeRecorded": true, "outcomeType": "TB_POSITIVE", "consentGiven": true }
}
```

HEALTH-SERVICES **health-tb-treatment**

DSTB / DRTB / TPT regimens with adherence tracking and adverse-event reporting.

verticalAttributes shape

```
{
  "vertical": "health-tb-treatment",
  "treatmentType": "DSTB",
  "regimenName": "2HRZE/4HR",
  "treatmentDurationDays": 180,
  "frequency": "Daily DOT",
  "monitoring": ["Sputum culture", "Liver function"],
  "adherenceTracking": true,
  "adverseEventReporting": true
}
```

Full Beckn message

```
{
  "context": { "domain": "health-services", "action": "on_confirm" },
  "itemAttributes": {
    "subCategory": { "code": "tb-treatment", "name": "TB Treatment" },
    "serviceName": "DSTB Regimen 2HRZE/4HR", "serviceMode": "FACILITY_VISIT",
    "duration": "P180D", "frequency": "RECURRING",
    "verticalAttributes": { "vertical": "health-tb-treatment", "treatmentType": "DSTB", "regimenName":
"2HRZE/4HR", "treatmentDurationDays": 180 }
  },
  "offerAttributes": { "priceAmount": 0, "pricingModel": "FREE", "isFreeAtPointOfCare": true, "coveredBy":
"GOVERNMENT_PROGRAM" },
  "fulfillmentAttributes": { "seekerLocation": { "county": "Nairobi", "subCounty": "Kibra" },
"providerCredentialRequired": "Clinical Officer", "deliveryMode": "FACILITY_VISIT", "serviceStatus": "RECORDED",
"followUpRequired": true },
  "orderAttributes": { "seekerUuid": "TIBU-PAT-554321", "outcomeRecorded": true, "outcomeType":
"TREATMENT_COMPLETED", "adverseEventReported": true, "adverseEventDetails": { "eventType": "Hepatotoxicity",
"severity": "MODERATE", "action": "Regimen modified", "outcome": "Resolved" } }
}
```

HEALTH-SERVICES **health-tb-supervision**

SCTLC / CTLC supervision visits, data-quality audits, mortality audits, mentorship. Seeker is the facility / programme; provider is the supervisor.

verticalAttributes shape

```
{
  "vertical": "health-tb-supervision",
  "supervisionType": "SCTLC_VISIT",
  "targetEntity": "FACILITY",
  "frequency": "Quarterly",
  "checklistTemplate": "https://ntld.go.ke/checklists/sctlc-v3.pdf",
  "includesPaymentClaim": true
}
```

Full Beckn message

```
{
  "context": { "domain": "health-services", "action": "on_confirm" },
  "itemAttributes": {
    "subCategory": { "code": "tb-supervision", "name": "TB Supervision" },
    "serviceName": "Quarterly SCTLC Supervision", "serviceMode": "FACILITY_VISIT",
    "frequency": "RECURRING",
    "verticalAttributes": { "vertical": "health-tb-supervision", "supervisionType": "SCTLC_VISIT", "targetEntity":
"FACILITY", "frequency": "Quarterly", "includesPaymentClaim": true }
  },
  "offerAttributes": { "priceAmount": 4000, "currency": "KES", "priceUnit": "per visit", "coveredBy":
"GOVERNMENT_PROGRAM" },
  "fulfillmentAttributes": { "facilityType": "Health Centre", "providerCredentialRequired": "SCTLC",
"deliveryMode": "FACILITY_VISIT", "serviceStatus": "RECORDED" },
  "orderAttributes": { "providerCredentialId": "SCTLC-NBI-009", "recordReference": "supervision-report-2026Q2-
001", "providerPaymentReceived": true, "providerAmountReceived": 4000 }
}
```

HEALTH-SERVICES **`advisory-health-tb`**

TB awareness, adherence education, side-effects management content. Discover-only.

verticalAttributes shape

```
{
  "vertical": "advisory-health-tb",
  "topic": "TB Awareness",
  "subTopic": "Adherence importance",
  "targetAudience": "Recently diagnosed TB patients",
  "sourceProvider": "Google Content API"
}
```

Full Beckn message (on_discover)

```
{
  "context": { "domain": "health-services", "action": "on_discover" },
  "itemAttributes": {
    "subCategory": { "code": "post-tb-care", "name": "Post-TB Care" },
    "serviceName": "Why TB adherence matters", "serviceMode": "DIGITAL",
    "contentType": "VIDEO", "contentUrl": "https://cdn.example/tb-adherence.mp4", "contentDuration": "PT4M30S",
    "targetBeneficiary": "Recently diagnosed TB patients",
    "availableLanguages": ["en", "sw"],
    "verticalAttributes": { "vertical": "advisory-health-tb", "topic": "TB Awareness", "subTopic": "Adherence
importance", "sourceProvider": "Google Content API" }
  },
  "offerAttributes": { "priceAmount": 0, "pricingModel": "FREE", "isFreeAtPointOfCare": true },
  "fulfillmentAttributes": { "deliveryMode": "DIGITAL", "serviceStatus": "DELIVERED" }
}
```

7. Beckn Core Entity Mapping

The four generic extension containers attach to the same set of Beckn v2 core entities regardless of domain or vertical. Only the contents differ.

EXTENSION CONTAINER	BECKN V2 CORE ENTITY	WHAT IT CARRIES
`itemAttributes`	Resource / Item	`ServiceItemAttributes` — what is being offered or sought
`offerAttributes`	Offer	`ServiceOfferAttributes` — pricing, coverage, availability, cancellation
`fulfillmentAttributes`	Performance / Fulfillment	`ServiceFulfillmentAttributes` — location, scheduling, delivery, lifecycle state
`orderAttributes`	Contract / Order	`ServiceOrderAttributes` — payment, feedback, outcome, consent, adverse events

> **Stable Beckn surface.** Onboarding new verticals never requires a new Beckn container or a new extension schema. The four generic schemas remain stable; the application layer reads `verticalAttributes.vertical` to dispatch vertical-specific logic.

8. Adding a New Vertical

Suppose you want to add an `advisory-finance` vertical (financial literacy content) under the `local-services` domain. No schema change is required. The steps are:

1. **Pick the Beckn domain.** Here: `local-services` (existing rural domain).
2. **Pick the category code.** Here: `Advisory-Content`.
3. **Pick a vertical identifier.** Convention: `<area>-<sub-area>`. Here: `advisory-finance`.
4. **Document the `verticalAttributes` shape.** Publish it as an extension of this cookbook:

```
{
  "vertical": "advisory-finance",
  "topic": "Personal Finance",
  "subTopic": "Savings Groups",
  "sourceProvider": "Kuza Finance Academy"
}
```

5. **Wire the application layer.** When the application sees `verticalAttributes.vertical === "advisory-finance"`, it routes to the finance-vertical handler. No Beckn protocol change, no OpenAPI change.

> **No schema change = no breaking change.** Existing integrations keep working. Only consumers that want to read / write the new vertical's fields need updates.

9. Domain Selection Guide

Use this guide to pick the right Beckn domain + vertical for a use case.

USE CASE	BECKN DOMAIN	VERTICAL
Farmer books a vet for poultry vaccination	`local-services`	`agri-livelihood-poultry`
Farmer orders feed for delivery	`local-services`	`agri-livelihood-feed`
Household pays school fees	`local-services`	`household-education`
Household seeks a general checkup	`local-services`	`household-health`
Farmer sells eggs to aggregator	`local-services`	`market-livestock-products`

USE CASE	BECKN DOMAIN	VERTICAL
Farmer receives agri advisory content	`local-services`	`advisory-agri`
Patient gets TB screening (sputum / GeneXpert)	`health-services`	`health-tb-screening`
Patient on DSTB / DRTB / TPT regimen	`health-services`	`health-tb-treatment`
SCTLC / CTLC visits facility for supervision	`health-services`	`health-tb-supervision`
Patient receives TB awareness / adherence content	`health-services`	`advisory-health-tb`

> **Rule of thumb.** If the use case is part of a disease-specific programme with clinical records and programme funding (NTLD, Global Fund, etc.), use `health-services`. Otherwise (general household health, livelihood, market, general advisory) use `local-services`.

10. Full OpenAPI 3.1 Bundle

Complete component block suitable for inclusion in the Beckn domain specification. All four generic schemas plus sub-schemas; usable for both `local-services` and `health-services` domains.

```yaml
# ============================================================
# NFH-Kuza-Google Extension Schemas (Domain-Agnostic)
# Domains: local-services | health-services | any future
# Protocol: Beckn v2.0.0
# ============================================================

openapi: "3.1.0"
info:
  title: NFH-Kuza-Google Generic Extension Schemas
  version: "3.0.0"
  description: >
    Four generic, domain-agnostic extension schemas for the
    Beckn v2 protocol, with a verticalAttributes extension bag
    for vertical-specific data.

components:
  schemas:

    # ---- Sub-schemas ----

    QuantitySpec:
      type: object
      required: [value, unit]
      properties:
        value: { type: number }
        unit: { type: string }
        description: { type: string }

    ServiceArea:
      type: object
      required: [countyCode, countyName]
      properties:
        countyCode: { type: string }
        countyName: { type: string }
        subCounties:
          type: array
          items:
            type: object
            required: [subCountyCode, subCountyName]
            properties:
```

```yaml
          subCountyCode: { type: string }
          subCountyName: { type: string }

    SeekerLocation:
      type: object
      required: [county, subCounty]
      properties:
        county: { type: string }
        subCounty: { type: string }
        ward: { type: string }
        village: { type: string }
        gpsCoordinates:
          type: object
          properties:
            latitude: { type: number }
            longitude: { type: number }

    Location:
      type: object
      properties:
        name: { type: string }
        address: { type: string }
        county: { type: string }
        subCounty: { type: string }
        gpsCoordinates:
          type: object
          properties:
            latitude: { type: number }
            longitude: { type: number }

    Feedback:
      type: object
      properties:
        rating:
          type: integer
          minimum: 1
          maximum: 5
        selectedOption: { type: string }
        additionalText: { type: string }

    # ---- VerticalAttributes (shared open bag) ----

    VerticalAttributes:
      type: object
      additionalProperties: true
      required: [vertical]
      properties:
        vertical: { type: string }

    # ---- 1. ServiceItemAttributes (itemAttributes) ----

    ServiceItemAttributes:
      type: object
      required: [subCategory, serviceName, serviceMode]
      properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        serviceName: { type: string }
        serviceDescription: { type: string }
        serviceMode:
          type: string
          enum: [ON_SITE_VISIT, FACILITY_VISIT, PROVIDER_LOCATION, COLLECTION_POINT, DELIVERY, DIGITAL, REMOTE]
        quantity: { $ref: "#/components/schemas/QuantitySpec" }
        duration: { type: string, format: duration }
```

```yaml
      contentType:
        type: string
        enum: [TEXT, VIDEO, PDF, AUDIO, IMAGE, INFOGRAPHIC, INTERACTIVE]
      contentUrl: { type: string, format: uri }
      contentMimeType: { type: string }
      contentDuration: { type: string, format: duration }
      targetBeneficiary: { type: string }
      documentsRequired: { type: array, items: { type: string } }
      frequency:
        type: string
        enum: [ONE_TIME, RECURRING, SEASONAL]
      requiredCapabilities: { type: array, items: { type: string } }
      availableLanguages: { type: array, items: { type: string } }
      tags: { type: array, items: { type: string } }
      verticalAttributes: { $ref: "#/components/schemas/VerticalAttributes" }

    # ---- 2. ServiceOfferAttributes (offerAttributes) ----

    ServiceOfferAttributes:
      type: object
      properties:
        priceAmount: { type: number }
        currency: { type: string }
        priceUnit: { type: string }
        pricingModel:
          type: string
          enum: [FREE, FIXED, PER_UNIT, PER_VISIT, NEGOTIABLE]
        isFreeAtPointOfCare: { type: boolean }
        coveredBy:
          type: string
          enum: [GOVERNMENT_PROGRAM, INSURANCE, OUT_OF_POCKET, NGO_FUNDED, GLOBAL_FUND, NOT_APPLICABLE]
        coverageNote: { type: string }
        acceptedPaymentMethods:
          type: array
          items: { type: string, enum: [CASH, MPESA, INSURANCE_CLAIM, NONE] }
        capacity:
          type: object
          properties:
            value: { type: number }
            unit: { type: string }
            period: { type: string }
        serviceAvailability:
          type: object
          properties:
            availableDays: { type: array, items: { type: string } }
            timeSlots: { type: array, items: { type: string } }
        cancellationPolicy:
          type: object
          properties:
            isCancellable: { type: boolean }
            windowHours: { type: integer }
            reasons: { type: array, items: { type: string } }
        validFrom: { type: string, format: date-time }
        validTo: { type: string, format: date-time }
        verticalAttributes: { $ref: "#/components/schemas/VerticalAttributes" }

    # ---- 3. ServiceFulfillmentAttributes (fulfillmentAttributes) ----

    ServiceFulfillmentAttributes:
      type: object
      required: [deliveryMode, serviceStatus]
      properties:
        serviceArea:
          type: array
          items: { $ref: "#/components/schemas/ServiceArea" }
        seekerLocation: { $ref: "#/components/schemas/SeekerLocation" }
        providerLocation: { $ref: "#/components/schemas/Location" }
        facilityType: { type: string }
```

```yaml
        requiredCapabilities: { type: array, items: { type: string } }
        providerCredentialRequired: { type: string }
        scheduledDate: { type: string, format: date }
        scheduledTimeSlot: { type: string }
        confirmedVisitTime: { type: string, format: date-time }
        actualServiceDate: { type: string, format: date }
        deliveryMode:
          type: string
          enum: [ON_SITE, FACILITY_VISIT, PROVIDER_LOCATION, COLLECTION_POINT, HOME_VISIT, TELE_HEALTH, MOBILE, DIGITAL]
        serviceStatus:
          type: string
          enum: [NEED_REGISTERED, SEARCHED, SELECTED, CONFIRMED, DELIVERED, RECORDED, OUTCOME_RECORDED, CANCELLED,
REJECTED, LOST_TO_FOLLOWUP, CLOSED]
        providerNotes: { type: string }
        followUpRequired: { type: boolean }
        referralRequired: { type: boolean }
        referralTarget:
          type: object
          properties:
            targetId: { type: string }
            targetName: { type: string }
            reason: { type: string }
            urgency: { type: string }
        verticalAttributes: { $ref: "#/components/schemas/VerticalAttributes" }

    # ---- 4. ServiceOrderAttributes (orderAttributes) ----

    ServiceOrderAttributes:
      type: object
      properties:
        seekerUuid: { type: string }
        providerCredentialId: { type: string }
        recordReference: { type: string }
        seekerPaymentMethod:
          type: string
          enum: [CASH, MPESA, INSURANCE_CLAIM, NONE]
        seekerPaymentStatus:
          type: string
          enum: [PAID, NOT_PAID, NOT_APPLICABLE]
        providerPaymentReceived: { type: boolean }
        providerAmountReceived: { type: number }
        paymentCurrency: { type: string }
        serviceConfirmedBySeeker: { type: boolean }
        providerFulfillmentStatus:
          type: string
          enum: [COMPLETED, PARTIAL, NOT_COMPLETED]
        outcomeRecorded: { type: boolean }
        outcomeType: { type: string }
        outcomeDate: { type: string, format: date }
        outcomeNote: { type: string }
        providerFeedback: { $ref: "#/components/schemas/Feedback" }
        seekerFeedback: { $ref: "#/components/schemas/Feedback" }
        followUpRequired: { type: boolean }
        followUpScheduledDate: { type: string, format: date }
        followUpNotes: { type: string }
        consentGiven: { type: boolean }
        consentTimestamp: { type: string, format: date-time }
        adverseEventReported: { type: boolean }
        adverseEventDetails:
          type: object
          properties:
            eventType: { type: string }
            severity: { type: string }
            onsetDate: { type: string, format: date }
            action: { type: string }
            outcome: { type: string }
        verticalAttributes: { $ref: "#/components/schemas/VerticalAttributes" }
```